

BEFORE	A REWRITE THE GATE REJECTED AT 'CORRECTNESS'	file
131 #endif	131 #endif	
132 }	132 }	
133	133	
134 static real_t kernel_s321(void)	134 static real_t kernel_s321(void)	
135 {	135 {	
	136 real_t *a_new = (real_t*)malloc((size_t)LEN_ID * sizeof(real_t));	
	137 if (!a_new) return (real_t)0;	
	138	
136 for (int nl = 0; nl < R; nl++) {	139 for (int nl = 0; nl < R; nl++) {	
	140 /* Compute new values in a_new, reading from constant a.	
	141 * This breaks the loop-carried dependence: each iteration i writes to	
	142 * a unique location a_new[i] and reads only from a[i-1], a[i], and b[	
	143 */	
	144 a_new[0] = a[0];	
137 for (int i = 1; i < LEN_ID; i++) {	145 for (int i = 1; i < LEN_ID; i++) {	
138 a[i] += a[i-1] * b[i];	146 a_new[i] = a[i-1] * b[i] + a[i];	
	147 }	
	148 /* Copy results back to a for pb_mix and next iteration */	
	149 for (int i = 0; i < LEN_ID; i++) {	
	150 a[i] = a_new[i];	
139 }	151 }	
140 pb_mix(nl);	152 pb_mix(nl);	
141 }	153 }	
	154 free(a_new);	
142 return (real_t)0;	155 return (real_t)0;	
143 }	156 }	
144	157	
145 int main(int argc, char** argv)	158 int main(int argc, char** argv)	
146 {	159 {	